

Yuliang Peng

ypeng12@umd.edu | [ypeng12.github.io/Mypage/](https://github.com/ypeng12) | [linkedin.com/in/ypeng12](https://www.linkedin.com/in/ypeng12) | 240-886-9716

EDUCATION

University of Maryland, College Park

- Master of Science in Applied Machine Learning
 - Bachelor of Science in Computer Science
 - Relevant Courses: Machine Learning, Computer Vision, Deep Learning, Data Structures and Algorithms
- Expected Graduation: May 2027
Graduated : May 2024

WORK EXPERIENCE / RESEARCH EXPERIENCE

University of Maryland, College Park - Research Assistant

Maryland | Nov 2025 - Present

- Designed and optimized **GPU-accelerated** spatiotemporal embeddings for high-rate **Event Camera** streams, enabling efficient ML inference via fused CUDA kernels and SIMD-optimized pipelines.
- Conducted data-driven analysis of aviation **network resilience** using network-level modeling on Delta system outage data.
- Applied **NLP** and **sentiment analysis** to TripAdvisor travel reviews to extract passenger perception dimensions

ModeSens - Software Engineer (Full-time)

Remote, USA | Apr 2025 - Sep 2025

- Built backend services for large-scale **product ingestion** from 30+ global retailers using **pipelines** and **serverless triggers**.
- Engineered production-grade data pipelines with **CI/CD** automation, reducing failures by **20%** and improving reliability.
- Designed large-scale **data ingestion systems** (Python, AWS S3) for feature extraction and search/recommendation tasks.
- Developed **multithreaded processing** components with concurrency control for high-throughput workloads.

Bank of China - Software Engineer (Full-time)

Los Angeles | July 2024 - Apr 2025

- Built backend services for financial transaction **monitoring systems**, supporting large-scale risk and compliance workflows.
- Engineered **data pipelines** for transaction processing and feature extraction, enabling analytics and fraud detection.
- Resolved **70+** production issues, reducing blocking incidents by **20%** and improving system stability

Air China - Software Engineer (Internship)

Chengdu, China | July 2024 - Apr 2025

- **Automated system backups** across Linux environments using Unix shell scripts, reducing manual workload by **25%**.
- Revamped the secure **login system** with mobile QR code scanning, streamlining user identification. (adopted in 2024)

PROJECT

Tabit - AI-Powered Chrome Extension | [Google Chrome Extension](#)

- Built an AI-powered Chrome extension using **semantic embeddings** for automatic tab grouping.
- Engineered a **real-time clustering pipeline** with Web Worker parallelization, supporting **100+** tabs with sub-**100ms latency**.
- Designed **incremental clustering** and **persistent session state** to enable efficient updates and seamless tab restoration.

AI Agent for Automated Data Processing | [Xfoundry](#)

- Built an **LLM-based agent** for automated multi-step data extraction and summarization from web sources.
- Designed **agent routing** and **tool-use pipelines** to orchestrate API calls across external services for complex task execution.
- Implemented **task planning** and **memory modules** to maintain state across multi-step interactions.

Cloud-Edge Collaborative Intelligent Agent Routing System | Google Hackathon

- Built a multi-level routing system for **edge and cloud LLM inference**, reducing **latency** and improving **time-to-first-token**.
- Developed **intent-based routing** to select between local models and external APIs, achieving **>95%** local processing.
- Implemented a post-processing framework to **mitigate hallucination** across multi-step tool calls, improving **response consistency** and system robustness.

Face Verification System | Machine Learning | PyTorch

- Built a face verification system using embeddings on the LFW dataset, enabling identity matching via distance thresholds.
- Designed a data pipeline for pair generation and **ROC-based threshold calibration**, improving accuracy and evaluation.

SKILLS

Programming Languages: C++, Python, C, Java, Dart, JavaScript, Unix Shell, SQL

Tools / Frameworks: AWS, Docker, Redis, NumPy, Pandas, Flask, Spring Boot, Vue.js, Flutter, Git, Sentry

Language: English (Fluent), Chinese (Native), Japanese (Native, N1)